

Evaluation Techniques

Evaluation

Evaluation is a vital part of the design process for interactive systems in human-computer interaction (HCI).

It involves assessing the effectiveness and usability of a system to identify potential issues and opportunities for improvement, with the goal of identifying potential problems and opportunities for improvement.

Goals of Evaluation

- **Assess System Functionality and Usability**- ensures it meets its goals and allows users to complete tasks efficiently. This is done through tests measuring user performance, helping identify and resolve any issues.
- **Assess Effect of Interface on User**- this means evaluating the overall user experience of the system. This feedback can be obtained through various means, such as interviews, questionnaires, and surveys.
- **Identify Problems Related to Functionality and Usability of Design**- this means evaluating the system in a holistic manner and can be achieved by combining the results of tests that assess system functionality and usability with feedback from users regarding their overall experience of using the system.

Evaluation Through Expert Analysis

Cognitive Walkthrough

- is an evaluation method used to assess the usability of a system
- a sequence of actions that a user would take to accomplish a task using the system
- The questions are:
 - Is the effect of the action the same as the user's goal at that point?
 - Will users see that the action is available?
 - Once users have found the correct action, will they know it is the one they need?
 - After the action is taken, will users understand the feedback they get?

Cognitive Walkthrough



- Understanding user goals – Does the interface clearly guide users toward their intended actions?
- Exploring different design choices – The image shows multiple variations of the remote, likely tested for usability.
- Predicting user difficulties – By analyzing these designs, developers could predict which layout would be easiest for users to understand and use.
- Iterative improvement – The walkthrough helps designers refine button placements, labels, and interactions based on potential user confusion.

Heuristic Evaluation

- is a method for evaluating a system's usability using a set of general heuristics, which are guidelines or principles.
- Multiple evaluators independently critique the system to find usability issues.
- is flexible, relatively cheap, and can be used on early design specifications, prototypes, and fully functioning systems.

Heuristic Evaluation Examples

1. Visibility of System Status

The UI should always keep users informed about what's happening.

Example: A loading indicator or progress bar for a file upload, showing the percentage completed.

2. Match Between System and the Real World

The design should speak the user's language with familiar concepts.

Example: An e-commerce checkout page using words like "Cart" and "Checkout," mimicking a real-world shopping experience.

3. User Control and Freedom

Users should have a way to undo or exit actions.

Example: A "Cancel" button during a form submission process or an "Undo" option after deleting an item.

Heuristic Evaluation (CTD)

4. Consistency and Standards

The design should follow platform conventions and ensure consistent behavior.

Example: A "hamburger menu" icon for navigation, commonly recognized across apps.

5. Error Prevention

The system should prevent errors before they happen.

Example: A warning message if users try to proceed without filling out required fields in a form

Review-Based Evaluation

Review-Based Evaluation involves evaluating the system based on feedback from a group of experts. This approach is useful for identifying usability issues that may not be immediately apparent to the designer.

Evaluation Through User Participation

Evaluation through user participation involves gathering feedback from users while they interact with the system. There are several styles of evaluation, including empirical methods, observational techniques, query techniques, and evaluation through monitoring physiological responses.

Empirical Methods: Experimental Evaluation

- Experimental evaluation tests systems under controlled conditions to study how design elements affect user behavior.
 - It identifies cause-and-effect relationships by testing hypotheses based on participant behavior.
 - Key factors include selecting appropriate participants, defining independent (manipulated) and dependent (measured) variables, and disproving the null hypothesis.
 - Careful experiment design ensures reliable and generalizable results.
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Empirical Methods: Experimental Evaluation

Example

A research team wants to evaluate a new search algorithm for an e-commerce website.

- Hypothesis: The new algorithm will help users find products 30% faster than the old one.
 - Experiment Setup: Two groups of users search for the same items—one using the old system and one using the new system.
 - Data Collection: Time taken to find products, number of clicks, user satisfaction ratings.
 - Conclusion: If users find items significantly faster with the new system, it supports the hypothesis.
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Observational Techniques: Think Aloud, Cooperative Evaluation, Protocol Analysis, Automated Analysis, Post-Task Walkthroughs

- Observational techniques- involve watching users interact with a system and collecting feedback to identify usability issues. Methods include Think Aloud, Cooperative Evaluation, Protocol Analysis, Automated Analysis, and Post-Task Walkthroughs.
 - Users are observed while performing tasks, with data recorded using paper notes, audio/video recording, or computer logs.
 - Each method has advantages and limitations, such as potential bias or altered user behavior due to observation.
 - provide valuable insights into interface problems that may not be obvious to designers.
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Observational Techniques Example

Scenario: Imagine a university is testing a new student portal for submitting assignments.

- The HCI research team observes students using the portal for the first time.
- Students are asked to "Think Aloud", verbalizing their thoughts while submitting an assignment.
- The researchers record errors, confusion, and frustrations using screen recording and notes.
- Later, they conduct a Post-Task Walkthrough, asking students what parts were difficult.

Through this process, the team finds that students struggle to locate the upload button. The university then redesigns the portal to improve clarity.

Query Techniques: Interviews, Questionnaires etc.

Query techniques involve gathering feedback from users using interviews, questionnaires, and surveys.

- Interviews are flexible and ideal for in-depth, high-level evaluations, allowing for follow-up questions based on responses.
 - Questionnaires are less flexible but can reach a larger group of participants and provide data that can be analyzed quantitatively.
 - Question types include general, open-ended, and scalar questions. Proper planning of both interview and questionnaire questions is essential for obtaining effective feedback.
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Evaluation through monitoring physiological responses

This discusses objective usability testing, which involves monitoring physiological responses of users to measure their cognitive processing and emotional response to computer use.

- Objective Usability Testing: Assesses cognitive and emotional responses during computer use.
 - Key Methods:
 - Eye Tracking: Identifies areas of difficulty by measuring eye movements.
 - Physiological Measurements: Tracks heart rate, sweat, muscle, and brain activity.
 - Insights: Indicates emotional states (e.g., stress, excitement) and user behavior.
 - Challenges: Requires more research on data interpretation and accessible equipment.
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Evaluation through monitoring physiological responses Example

Imagine a gaming company testing a new virtual reality (VR) game:

- Objective: To see how players feel while navigating through stressful in-game situations (e.g., facing enemies).
- Method:
 - Eye Tracking: Measures where players are looking during intense scenes to identify focus areas.
 - Physiological Measurements: Sensors track heart rate and sweat gland activity to detect excitement or anxiety.
- Insights:
 - Players' heart rates increase during certain game events, indicating stress.
 - Eye tracking shows players focus on the game's environment, suggesting immersion.
- Outcome: The company uses this data to adjust the game's difficulty, ensuring it is engaging without overwhelming players.

Thank You
